

Hybrid V2 Physics API

Calculator-first inference with Qwen2.5-3B LoRA adapters for numeric parsing, locked-trace explanation, and conceptual Type 1 / CHLT reasoning.

Main principle

Use LLMs for language-heavy work, but keep arithmetic inside deterministic physics formulas.

Numeric authority

Type 2 answers are recomputed by `physics_calculator_v2` after payload validation.

CHLT clarification

CHLT is Yes / No / Uncertain, but it may still require numeric calculation before choosing the label.

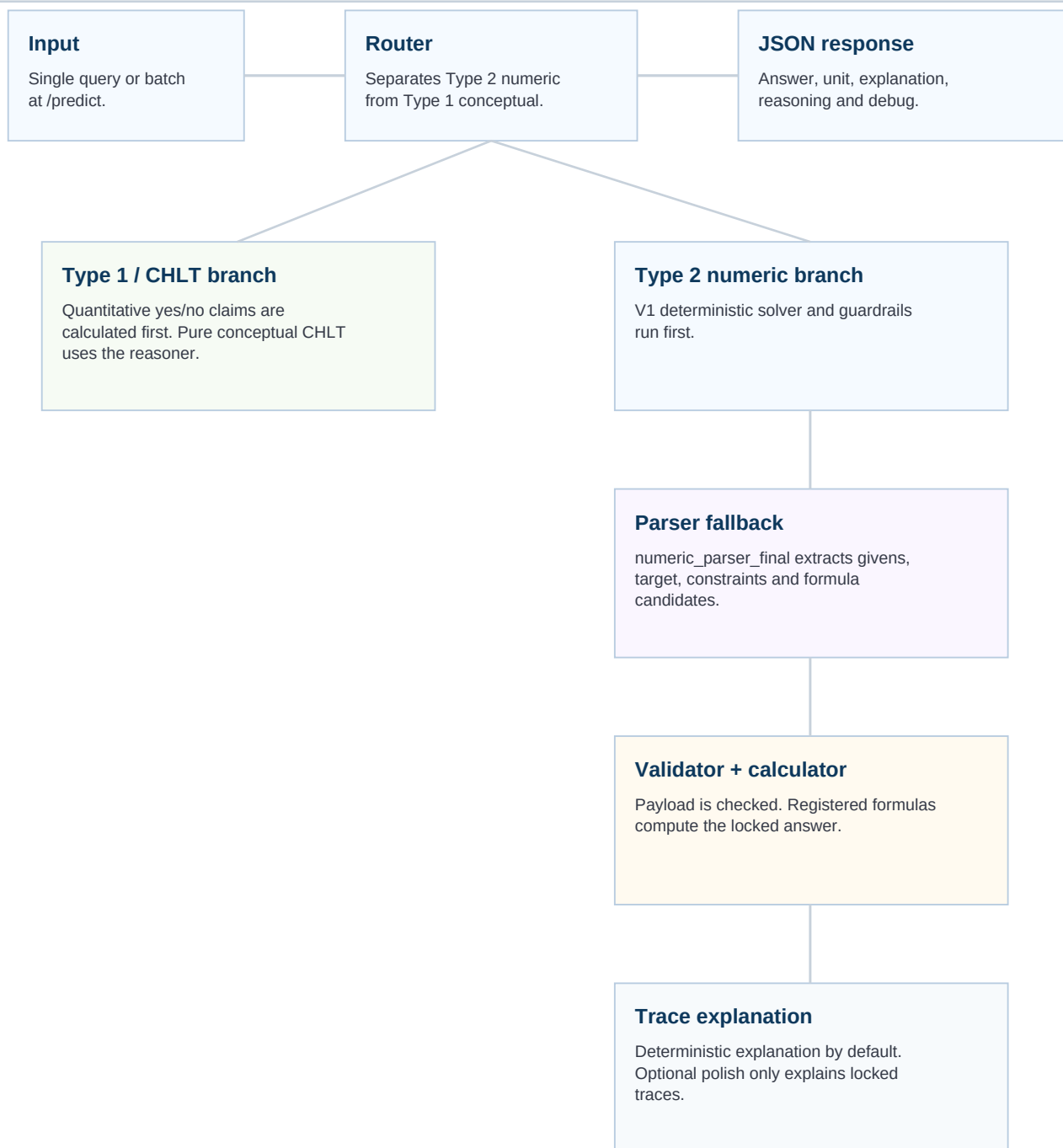
Final package

The runtime uses one 3B base model and three LoRA adapters: numeric parser, trace explainer, and conceptual reasoner.

Why this design

- Fast deterministic paths solve common numeric physics without waiting for vLLM.
- Unseen wording is handled by a parser adapter, then verified by calculator code.
- Explanation polish is optional and cannot change locked values.
- Conceptual questions are isolated from numeric solving.

Runtime Architecture



Routing guardrail

A numeric yes/no request should not be solved directly by the conceptual adapter. The API first computes the relevant quantity, compares it with the claim, and only then returns Yes, No, or Uncertain.

Adapters and Contracts

Each LoRA adapter has a narrow role and a JSON contract. This keeps the system auditable and reduces hallucinated outputs.

Adapter	Role	Rows	Contract
numeric_parser_final	Turns natural Type 2 wording into structured quantities and formulas.	1443 / 220	No answer, no Python, no final_result, no CoT.
trace_explainer_final	Explains a locked calculator trace in readable language.	1196 / 163	Explanation only; answer, unit and formula are immutable.
chlt_reasoner_final	Conceptual Type 1 reasoner. Pure CHLT subset is Yes / No / Uncertain.	334 / 73	Used after quantitative claim verification fails or when the prompt is purely conceptual.

Why no planner-code adapter

- Generated code can be syntactically invalid or use the wrong equation.
- Generated answer fields can disagree with the code result.
- The final design asks the model to parse only, then lets deterministic formulas calculate.

Numeric parser allowed output

topic, question_kind, target, givens, constraints, formula_candidates, confidence

Numeric parser forbidden output

answer, final_result, unit_answer, python_code, golden_code, chain-of-thought

Calculator and Verification

69 formulas

Registered branches in formula_bank.py and calculator.py.

Payload validator

Checks JSON, spans, roles and formula IDs before calculation.

Locked result

Final numeric answer is computed by deterministic code.

Covered formula families

Circuits

Ohm law, power, wire resistance, series/parallel networks.

Capacitors

Charge, voltage, capacitance, stored energy and equivalent capacitance.

LC / RLC

LC energy and frequency, resonance, reactance and impedance.

Electrostatics

Coulomb force, point-charge fields and vector resultants.

Induction

Magnetic flux, Faraday emf, solenoid field and inductor energy.

Measurement

Relative error, absolute error and propagated uncertainty.

Validation ladder

- 1. Parse JSON and required fields.
- 2. Reject forbidden generated fields in numeric parser output.
- 3. Check raw spans against the question.
- 4. Check formula candidates against the registered formula bank.
- 5. Compute and format the final answer with deterministic unit conversion.

Deployment and API Contract

Item	Final setting
Endpoint	/predict returns one JSON result per query.
Kaggle package	kaggle_api_package_hybrid_v2_clean.zip
Notebook	result/Notebook/deploy_physics_api_qwen3b_vllm_kaggle.ipynb
Runtime default	Numeric parser and conceptual reasoner enabled. Polish disabled by default for speed.
Health checks	/health and /v1/models expose pipeline and model status.

Response fields

- query_id: copied from the request.
- answer and unit: final formatted result.
- explanation: deterministic or locked-trace explanation.
- reasoning: concise steps/evidence for auditability.
- debug: optional timing, route, confidence and model status.

Final design claim

The system combines V1 deterministic speed with V2 parser-plus-calculator flexibility. The model helps understand and explain; the calculator owns numeric truth.